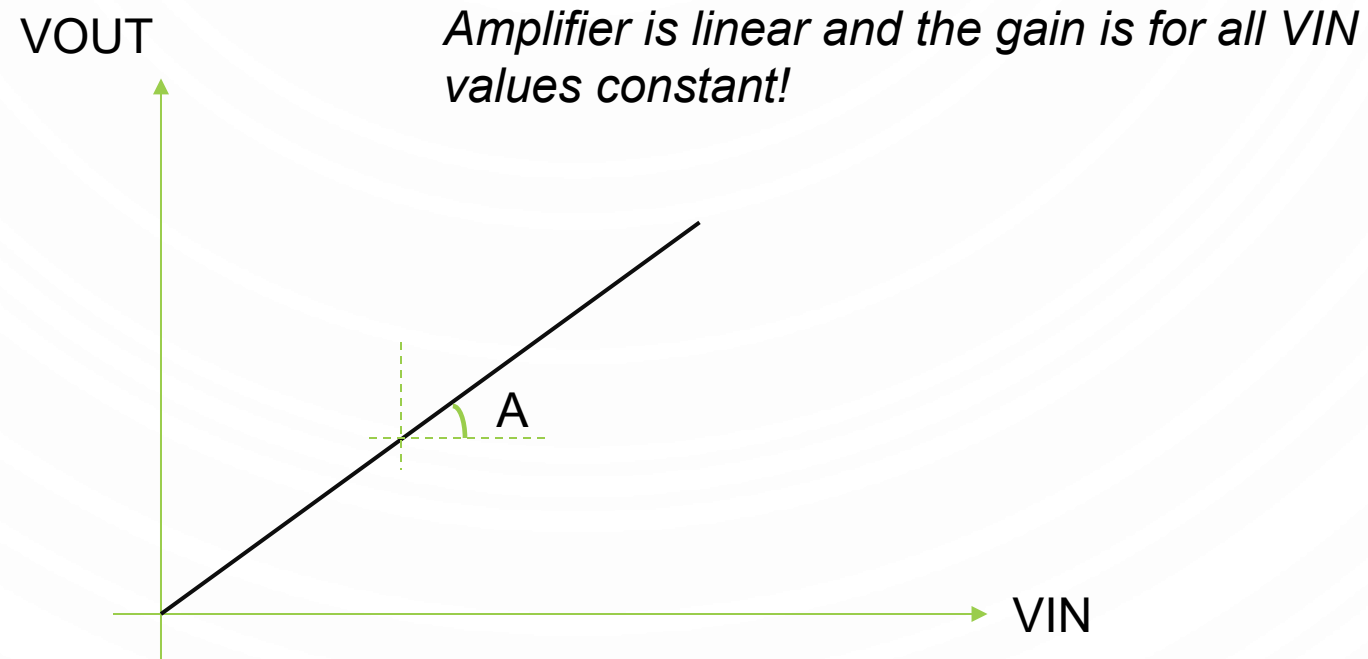
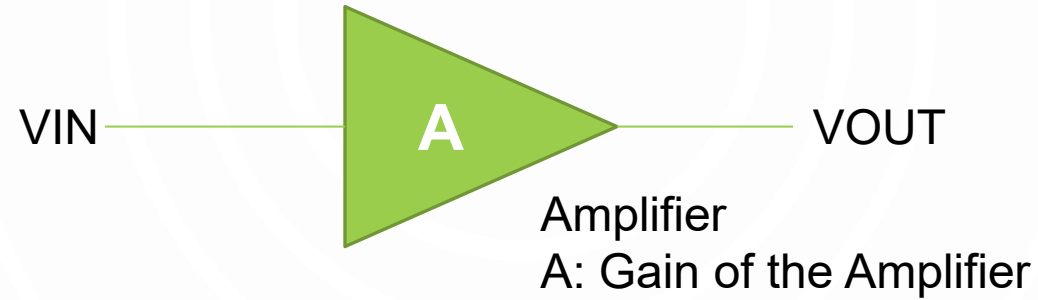
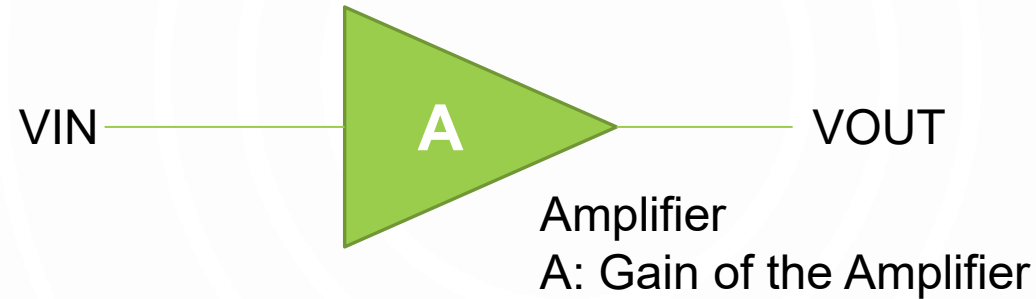


Single Stage Amplifiers

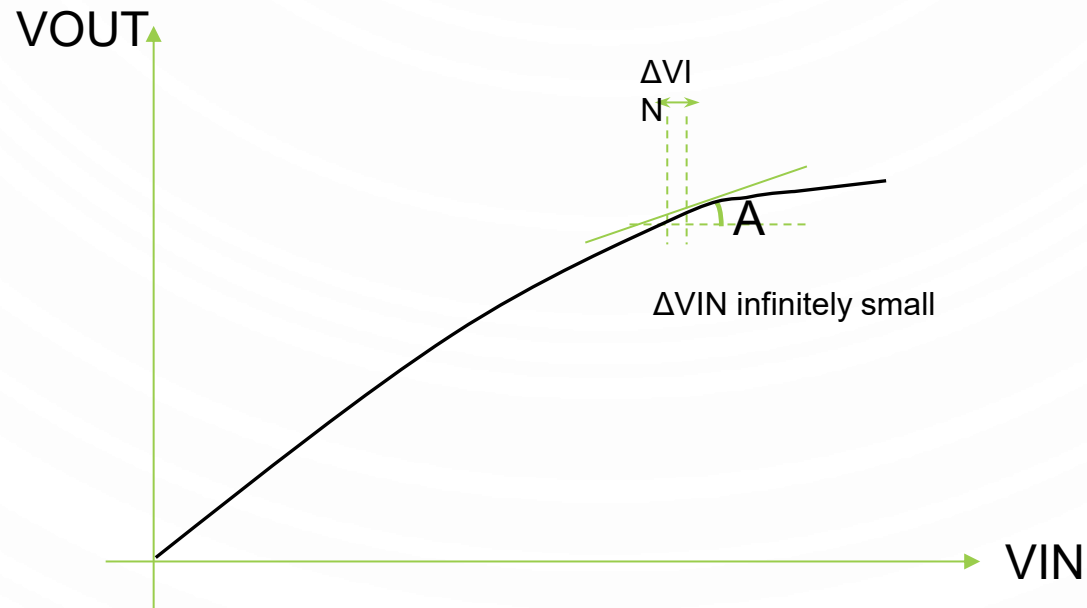
SMALL-SIGNAL AND LARGE-SIGNAL ANALYSIS



SMALL-SIGNAL AND LARGE-SIGNAL ANALYSIS



Amplifier is non-linear and the gain varies over V_{IN} ! Amplifier is assumed to be linear in an indefinitely small range! This is a small-signal approximation. If the analysis takes into account the change over different V_{IN} , this is a large-signal analysis and implies more complicated circuit models and analysis!!!



This plot is just an example and does not reflect the linearity of a real-world amplifier!

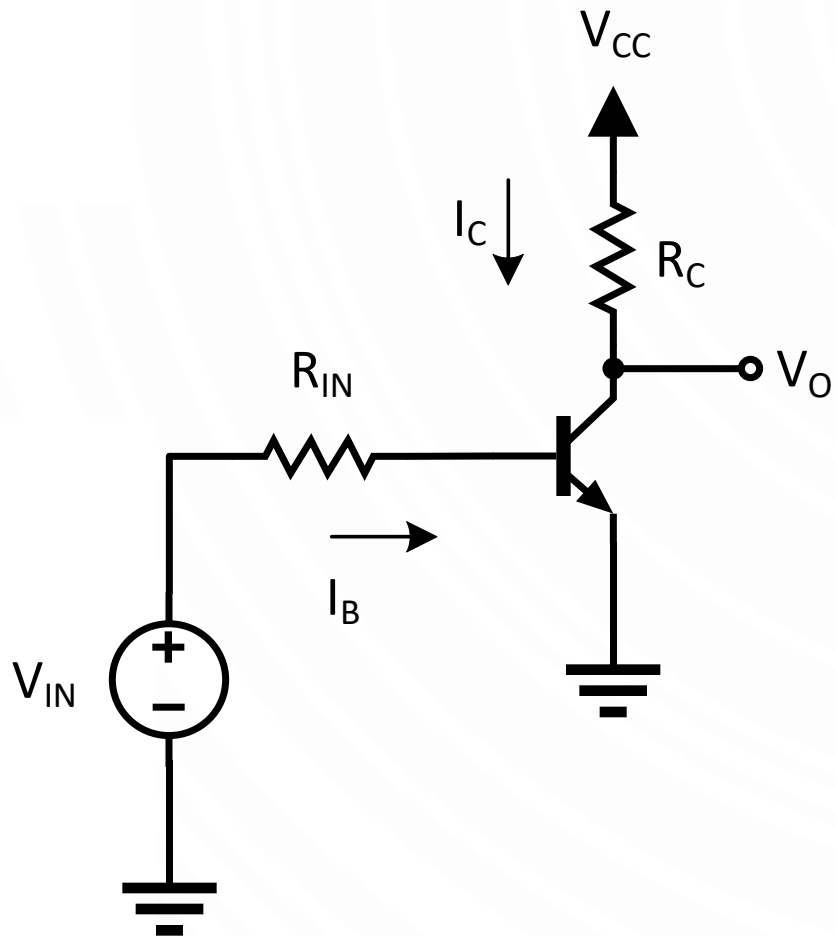
SMALL- AND LARGE-SIGNAL ANALYSIS

Why is an amplifier non-linear?

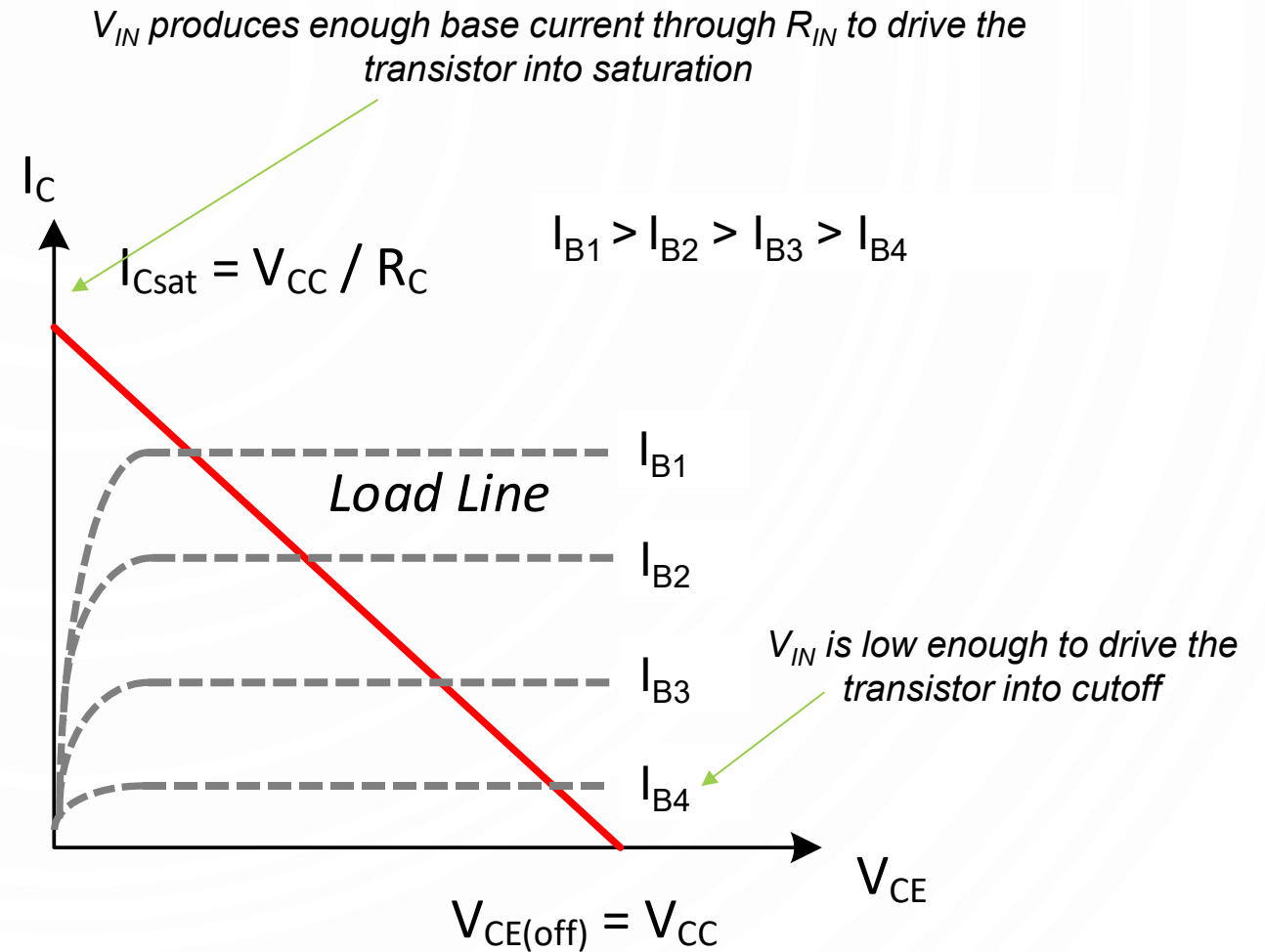
This will be explained in the next slides...

SIMPLE APPLICATION EXAMPLES

LARGE-SIGNAL ANALYSIS



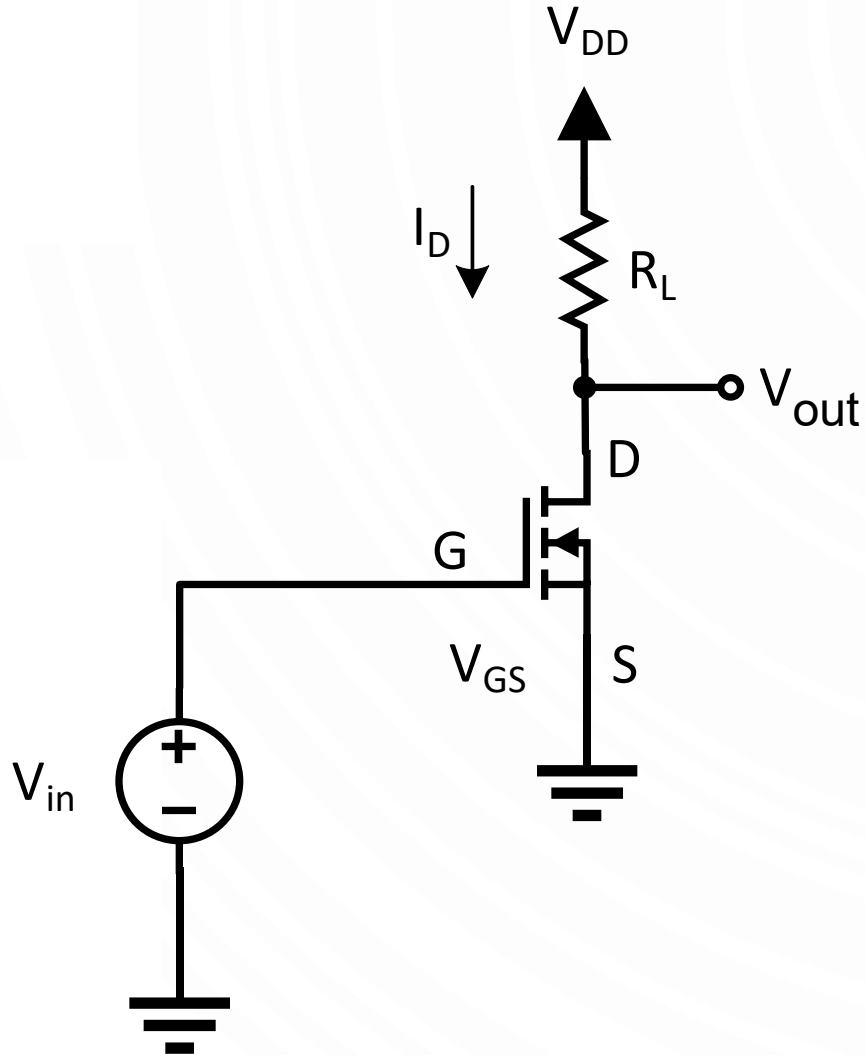
$$V_O = V_{CC} - R_C I_C$$



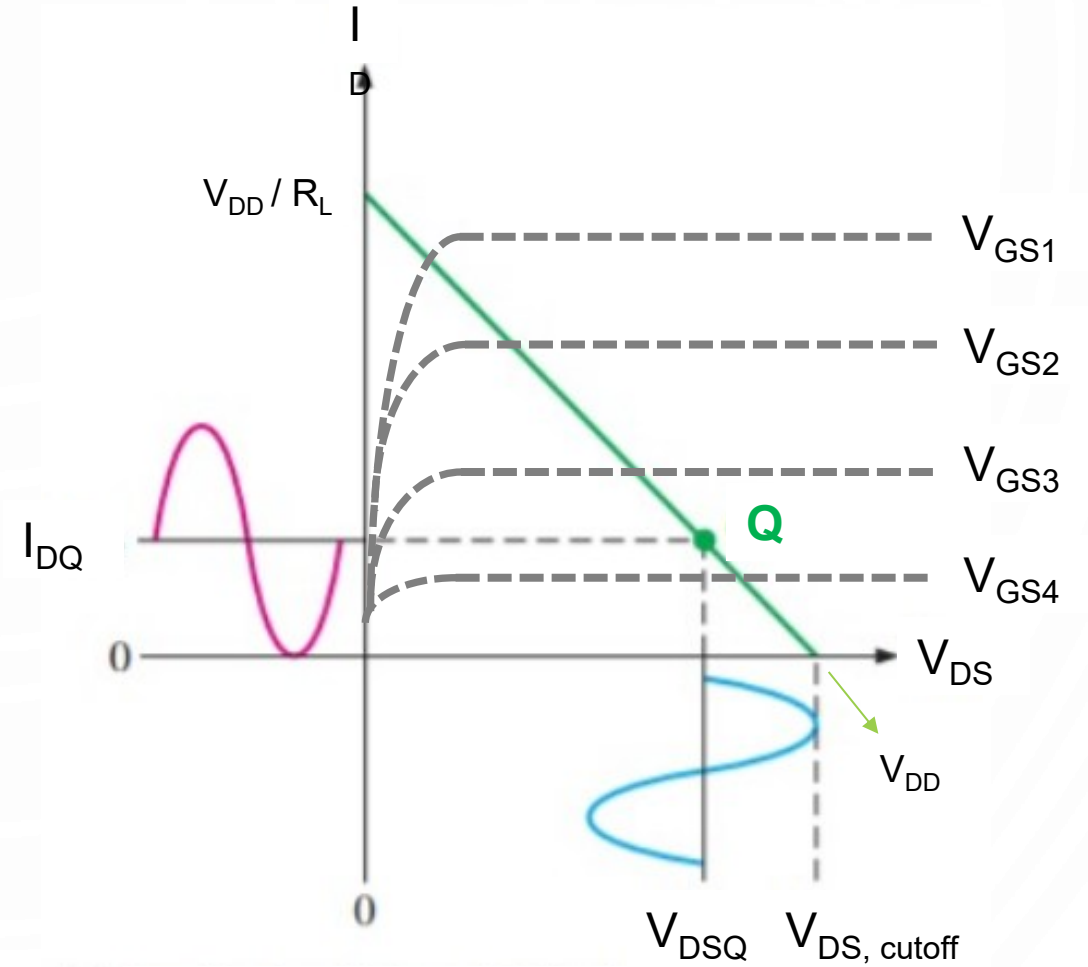
A similar load line can be derived also for MOSFETs

SIMPLE APPLICATION EXAMPLES

LARGE-SIGNAL ANALYSIS



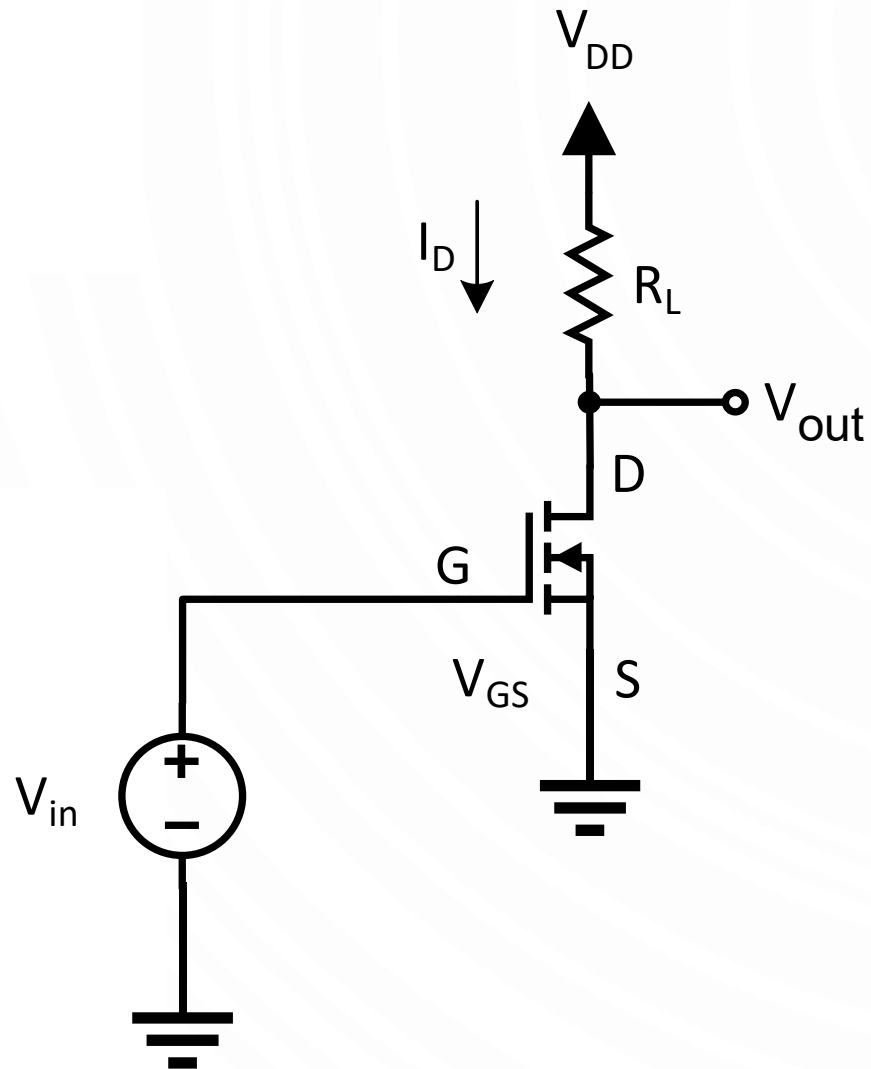
Common-source stage



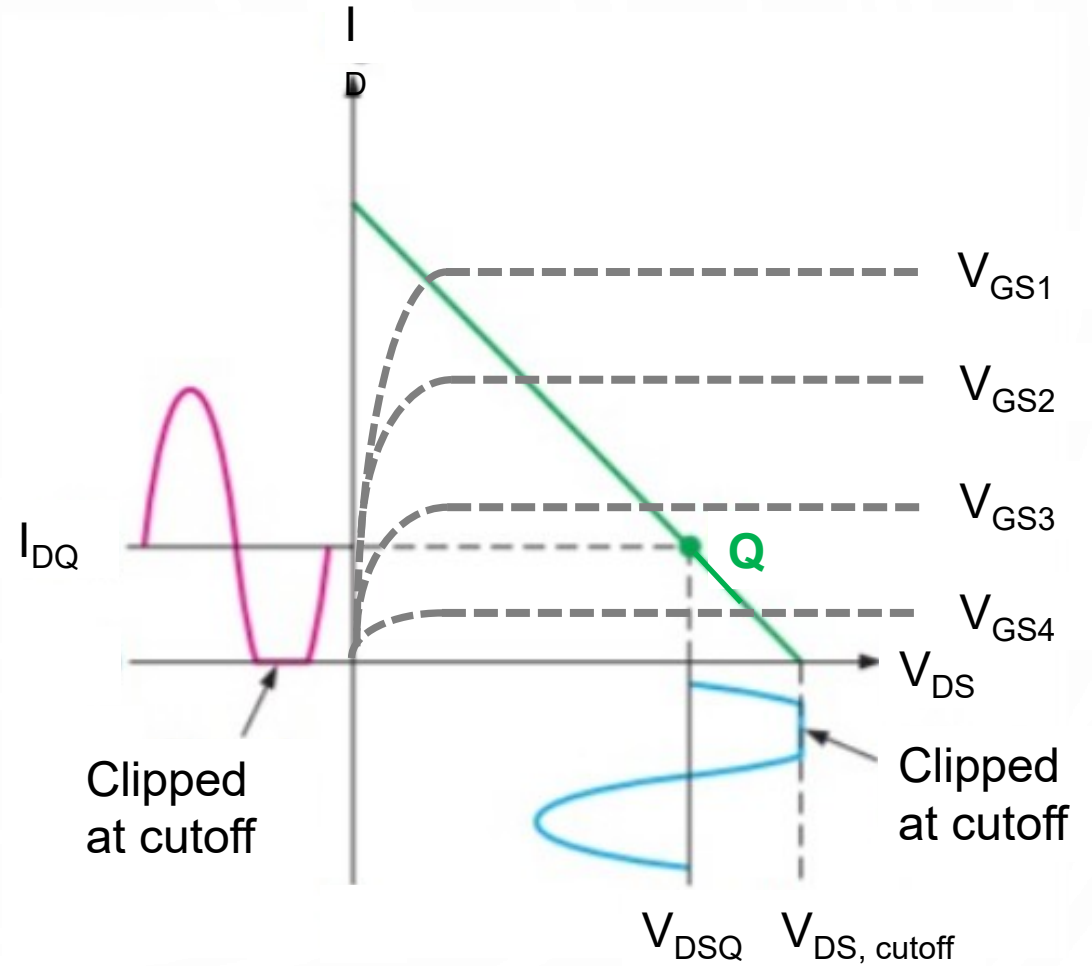
$$V_{GS1} > V_{GS2} > V_{GS3} > V_{GS4}$$

SIMPLE APPLICATION EXAMPLES

LARGE-SIGNAL ANALYSIS



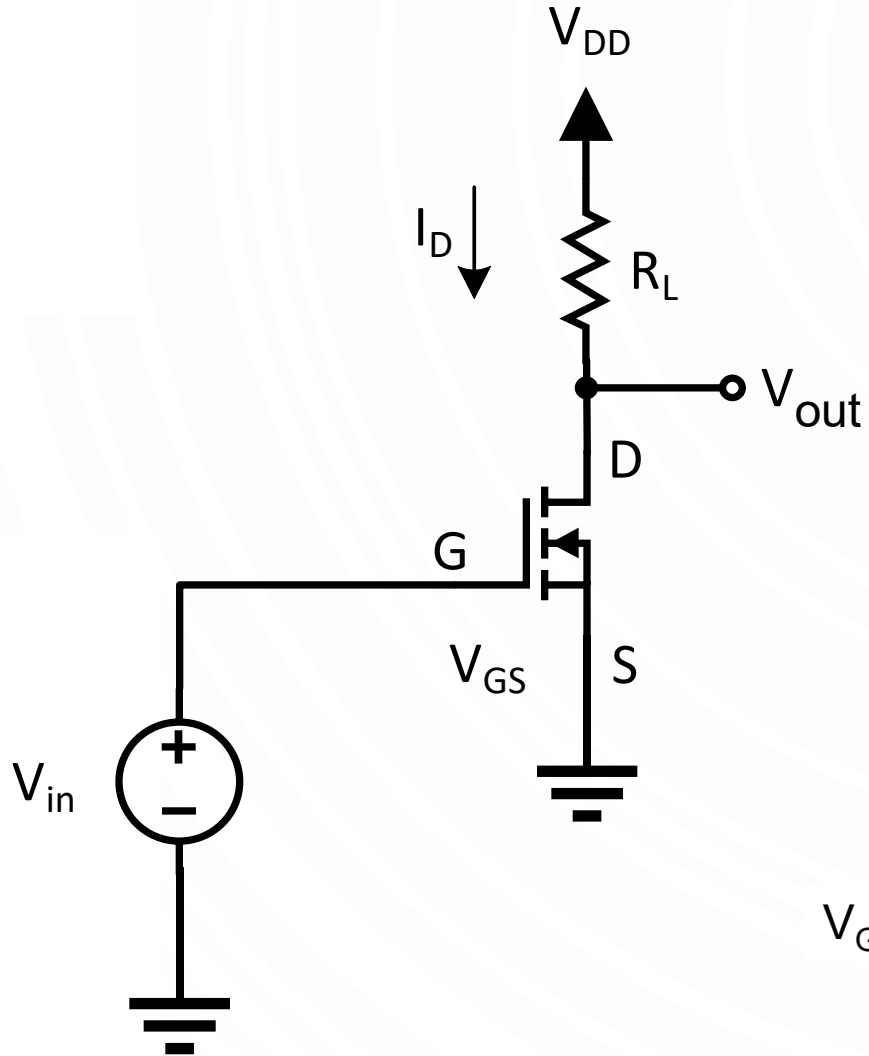
Common-source stage



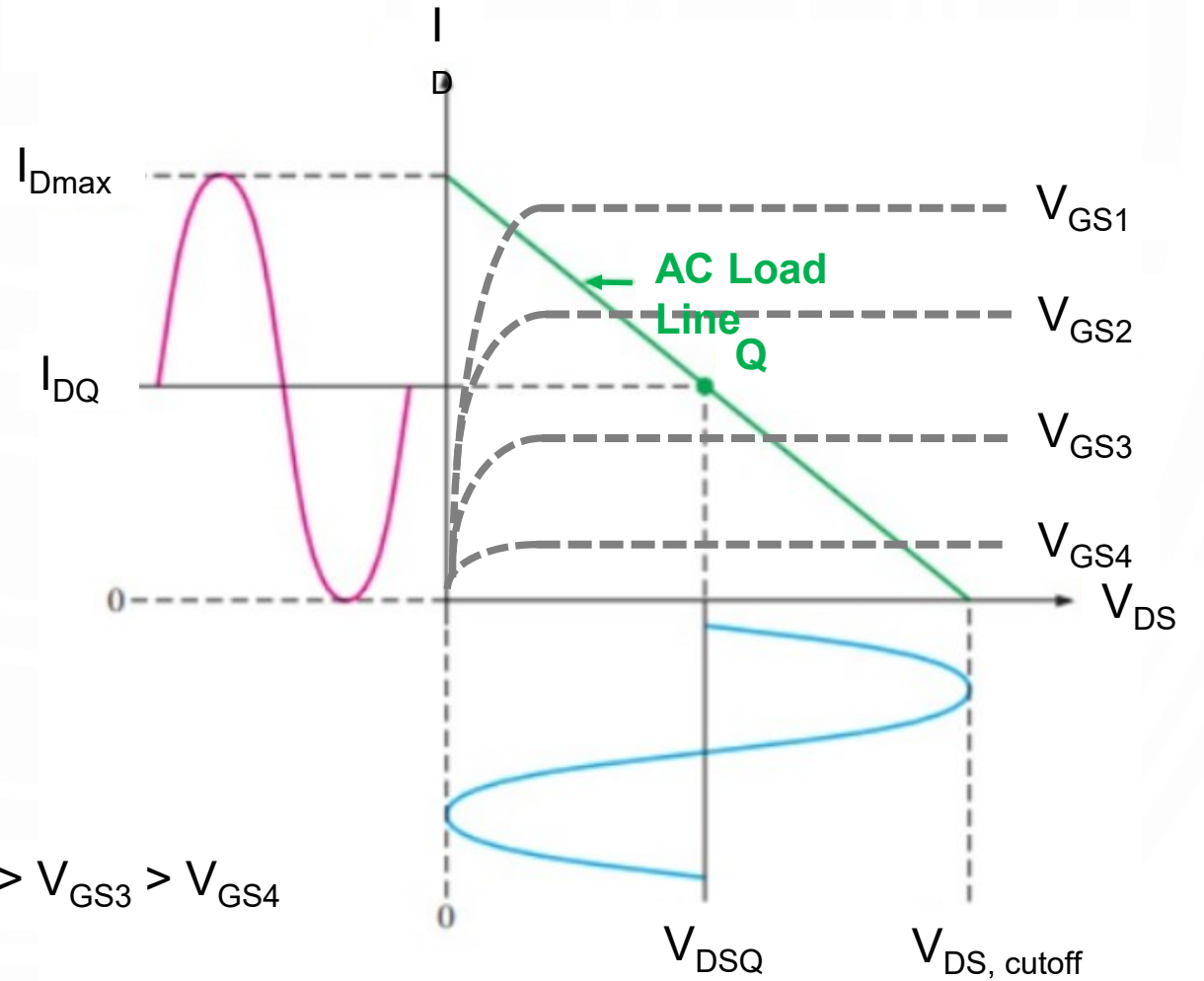
$$V_{GS1} > V_{GS2} > V_{GS3} > V_{GS4}$$

SIMPLE APPLICATION EXAMPLES

LARGE-SIGNAL ANALYSIS



Common-source stage



$$V_{GS1} > V_{GS2} > V_{GS3} > V_{GS4}$$

The quiescent point Q shall be in the middle of the load line for maximum signal output amplitude

SIMPLE APPLICATION EXAMPLES

LARGE-SIGNAL ANALYSIS

Consider also the following important notes when selecting the operating point of the amplifier:

- If V_{DS} enters the triode region:
 - The MOSFET behaves like a resistor, not a current source.
 - Gain decreases.
 - Output waveform compresses (clipping at the low-voltage side).
- If V_{GS} is small (low I_D):
 - g_m decreases and is not constant.
 - The transfer characteristic becomes strongly nonlinear.
 - Harmonic distortion increases.



see next page...

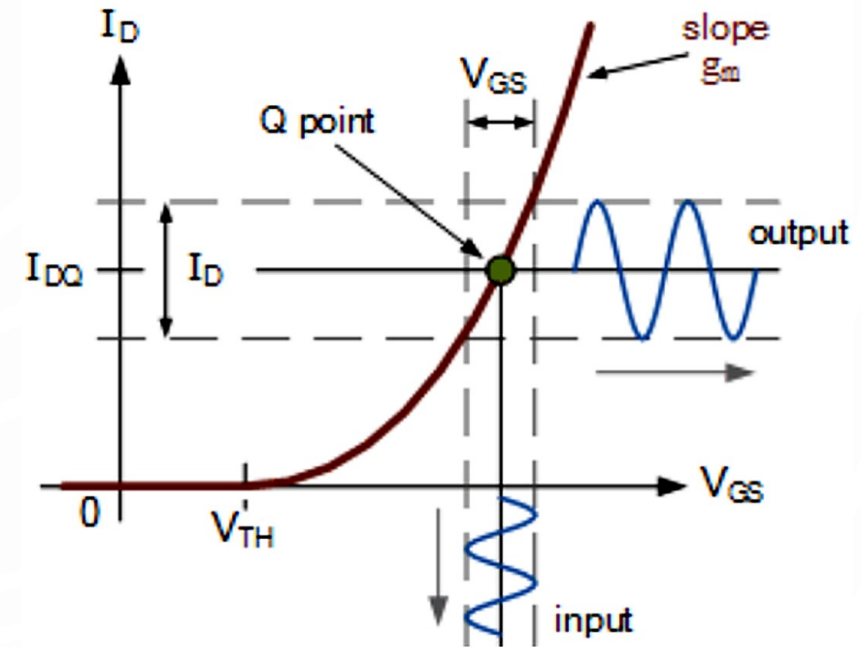
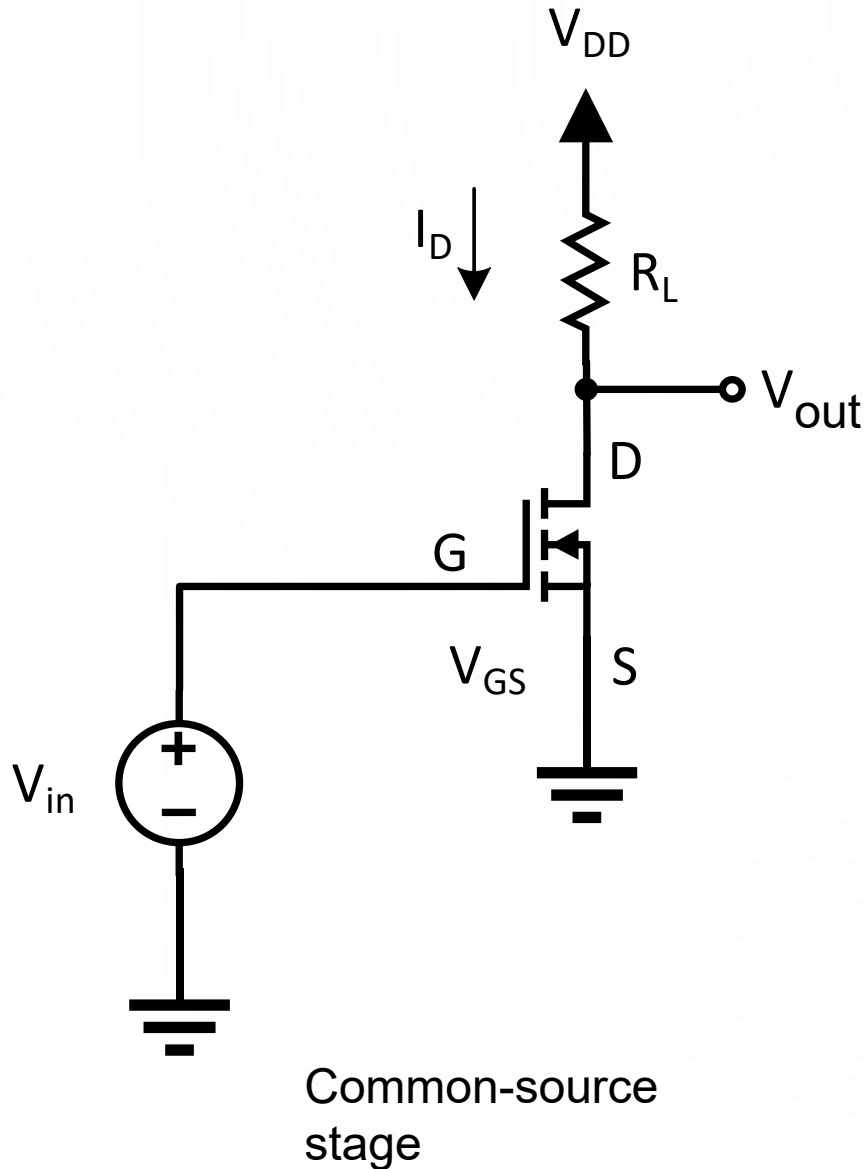
Conclusion:

Large V_{DS} swing → triode distortion.

Small V_{GS} (low current) → low-current distortion.

SIMPLE APPLICATION EXAMPLES

SMALL-SIGNAL ANALYSIS



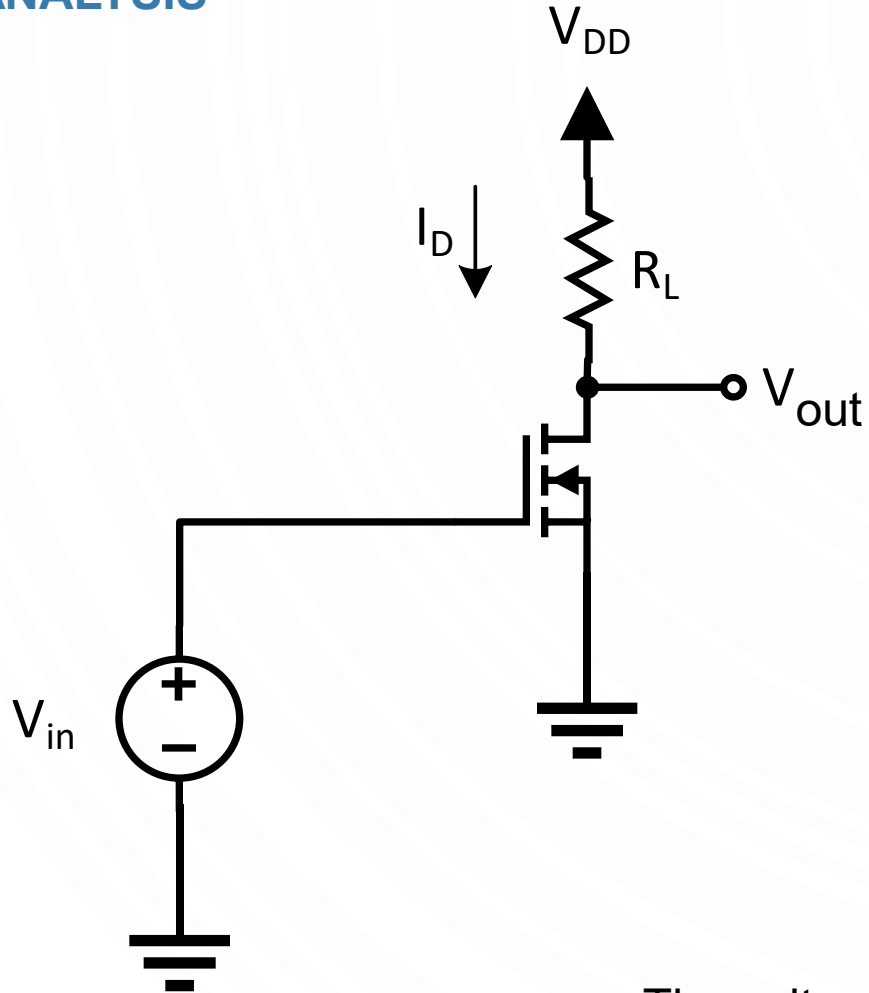
$$\frac{v_{out}}{v_{in}} = -g_m R_L$$

g_m : transconductance

behaves linear in a very small V_{GS} range

SINGLE STAGE AMPLIFIERS – COMMON SOURCE AMPLIFIER

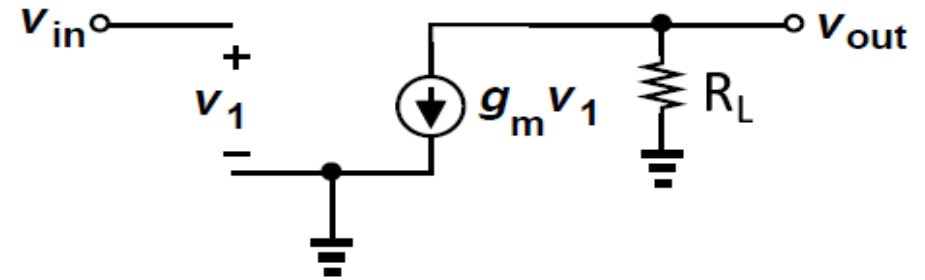
SMALL-SIGNAL ANALYSIS



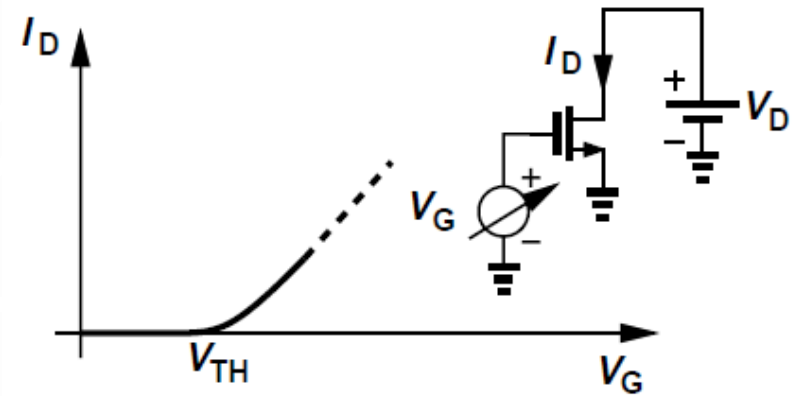
Common-source stage

The voltage gain is negative!

g_m : transconductance



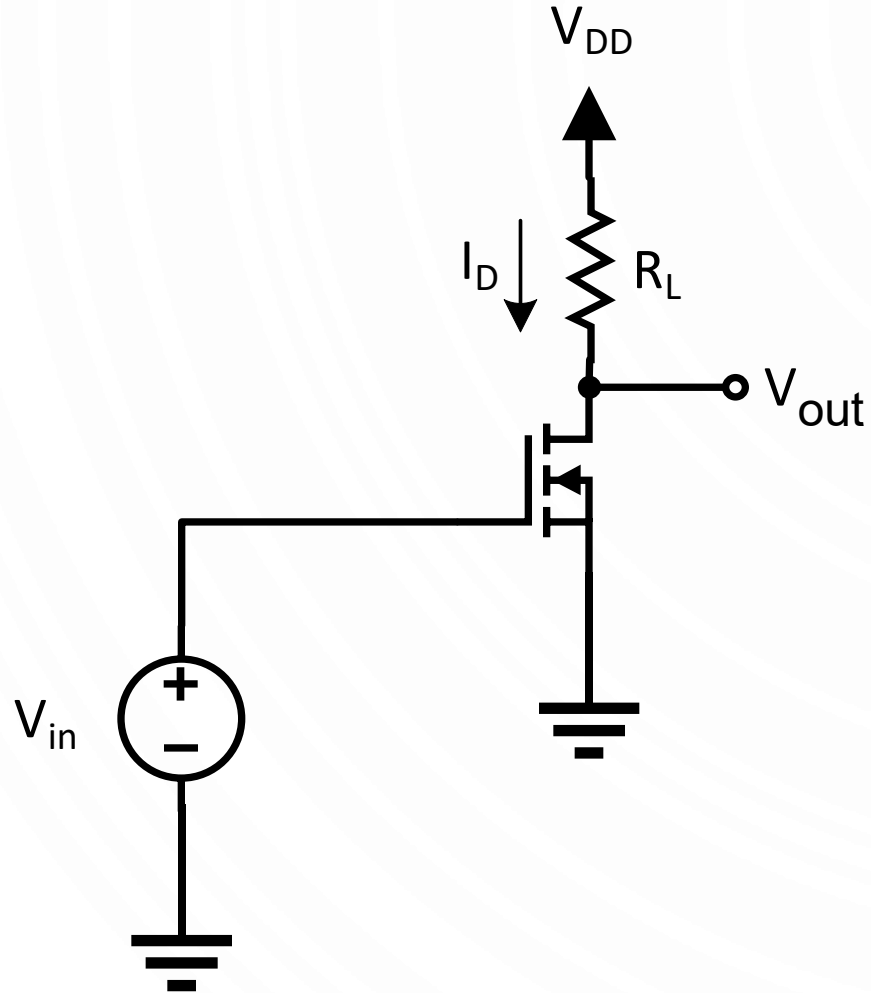
Simplified small-signal (AC) equivalent circuit of common-source stage (C_{GS} and R_{DS} neglected)



$$\frac{v_{out}}{v_{in}} = -g_m R_L$$

SINGLE STAGE AMPLIFIERS – COMMON SOURCE AMPLIFIER

SMALL-SIGNAL ANALYSIS



Common-source
stage

$$g_m = \frac{\partial I_D}{\partial V_{GS}}$$

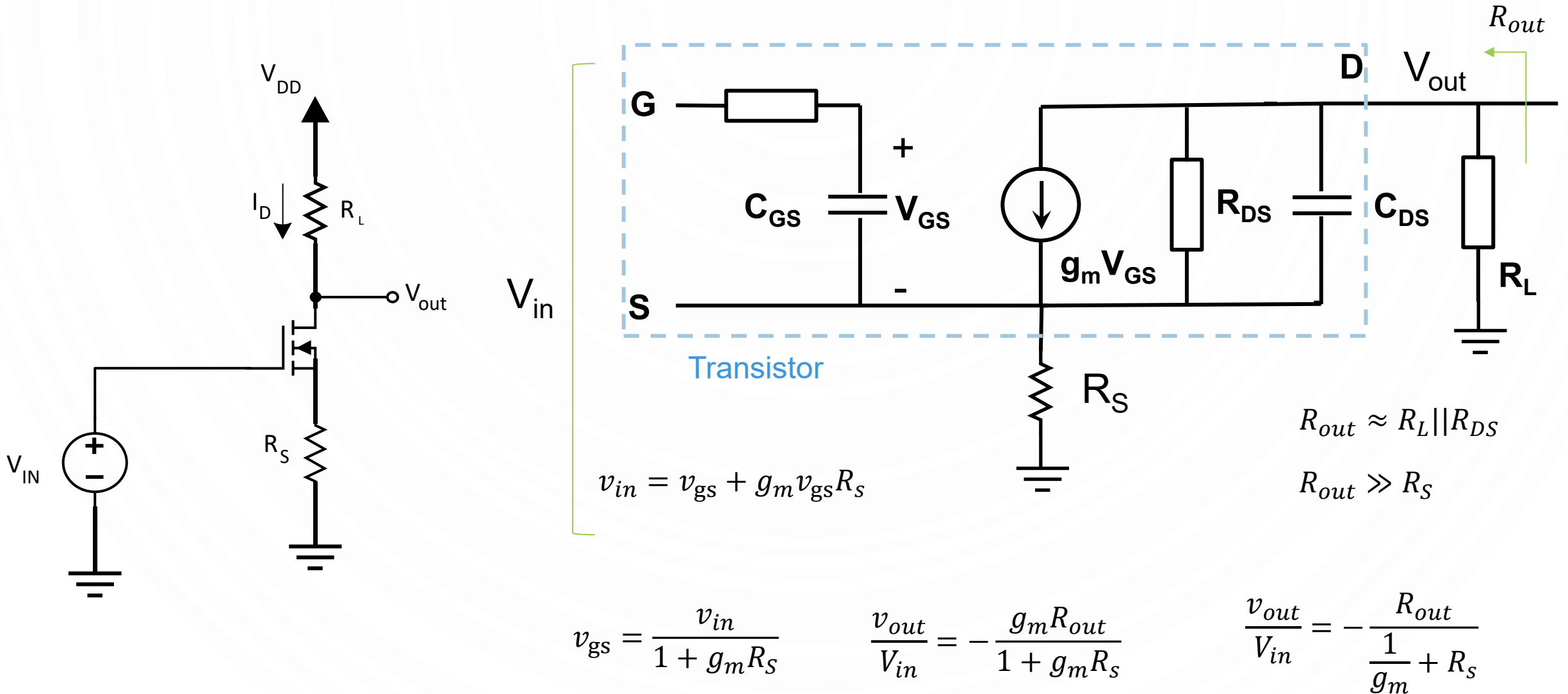
$$I_D = I_{Dsat} = \frac{W\mu_n C_{ox}}{2L} [(V_{GS} - V_T)^2] \quad V_{Dsat} \leq V_{DS}$$

$$g_m = \mu_n C_{ox} \frac{W}{L} (V_{GS} - V_{TH})$$

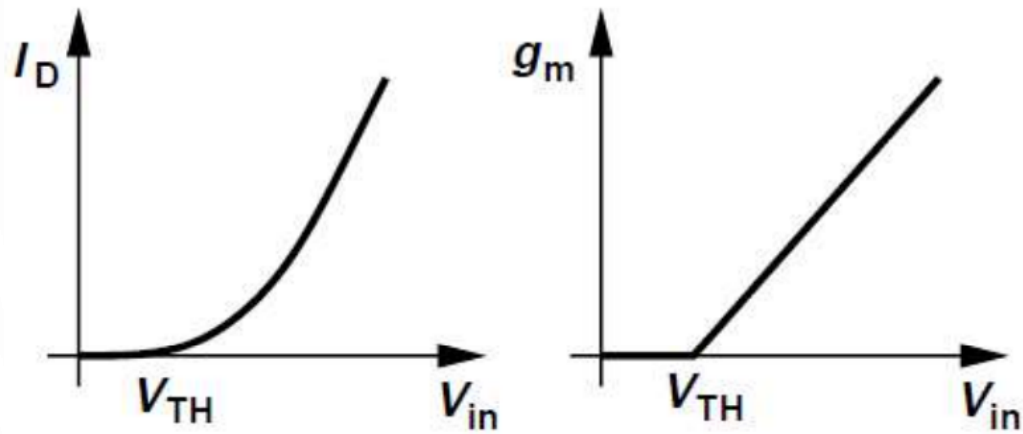
$$g_m = \sqrt{2\mu_n C_{ox} \frac{W}{L} I_D}$$

$$A_v = -g_m R_L$$

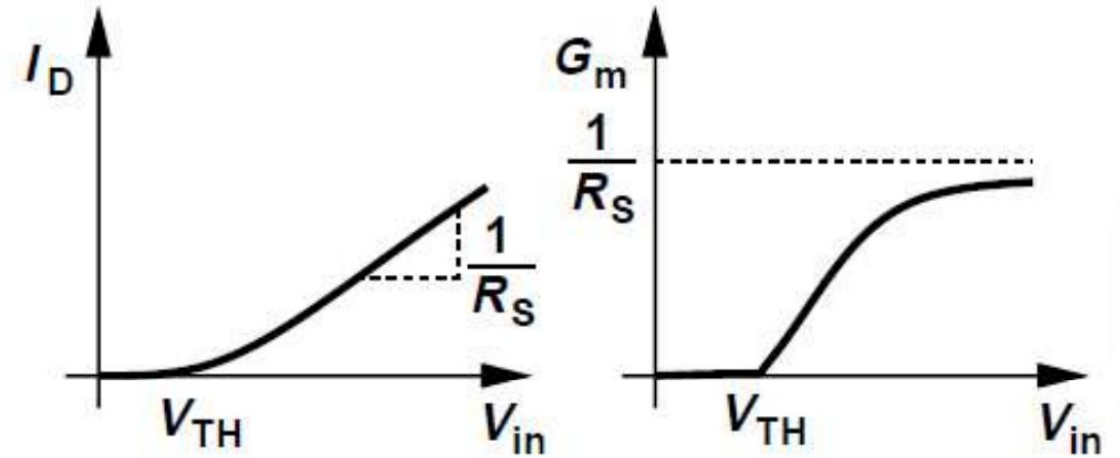
SINGLE STAGE AMPLIFIERS – COMMON SOURCE AMPLIFIER WITH SOURCE DEGENERATION



SINGLE STAGE AMPLIFIERS – COMMON SOURCE AMPLIFIER WITH SOURCE DEGENERATION



$$R_S = 0$$



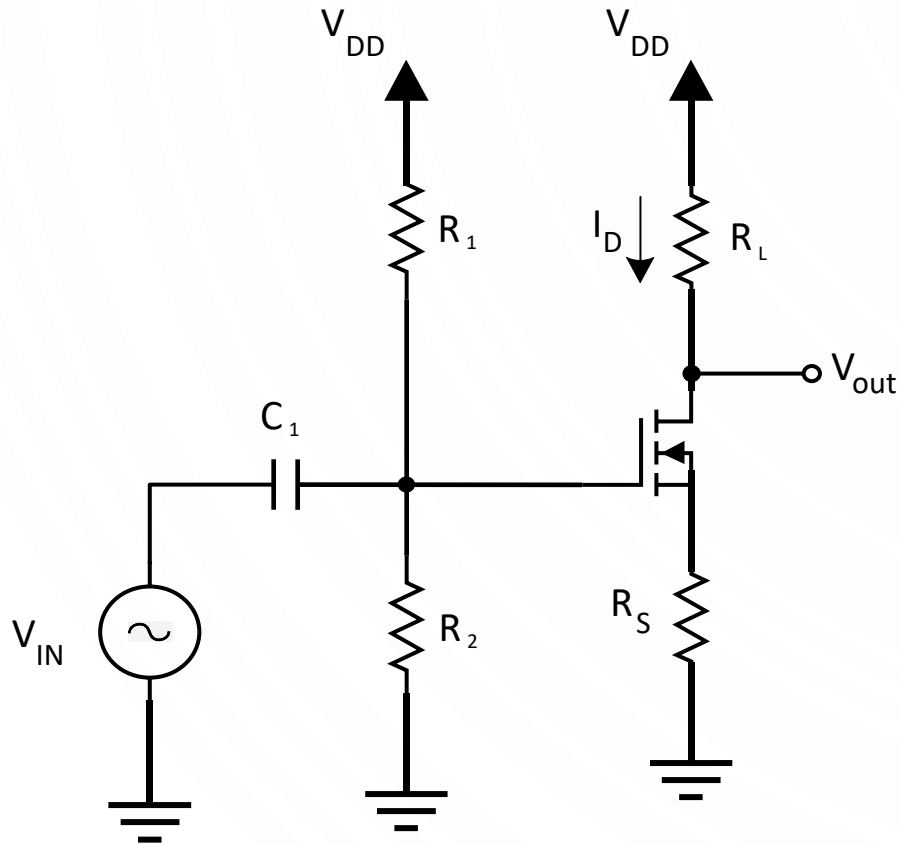
$$R_S \neq 0$$

I_D and g_m vary with V_{in} . At low current levels, turn-on behavior is similar to when $R_S = 0$,

since

$$\frac{1}{g_m} \gg R_S$$

SINGLE STAGE AMPLIFIERS – COMMON SOURCE AMPLIFIER BIASING



V_{IN} : AC voltage source

C_1 : AC coupling capacitor

If the input impedance of the MOSFET is assumed to be infinite, then

$$R_{in} = R_1 || R_2$$

Without R_1 and R_2 , the input impedance of the amplifier is very large!

In AC analysis, V_{DD} can be assumed to be connected to ground → why? usually, there is a decoupling capacitor between V_{DD} and ground which short-circuits the V_{DD} node to ground when making an AC analysis!

SINGLE STAGE AMPLIFIERS – INPUT AND OUTPUT RESISTANCES

How to calculate input resistance:

1. Open-circuit the output.
2. Apply a test voltage v_{in} at the input.
3. Compute the resulting input current i_{in} .
4. Take the ratio:

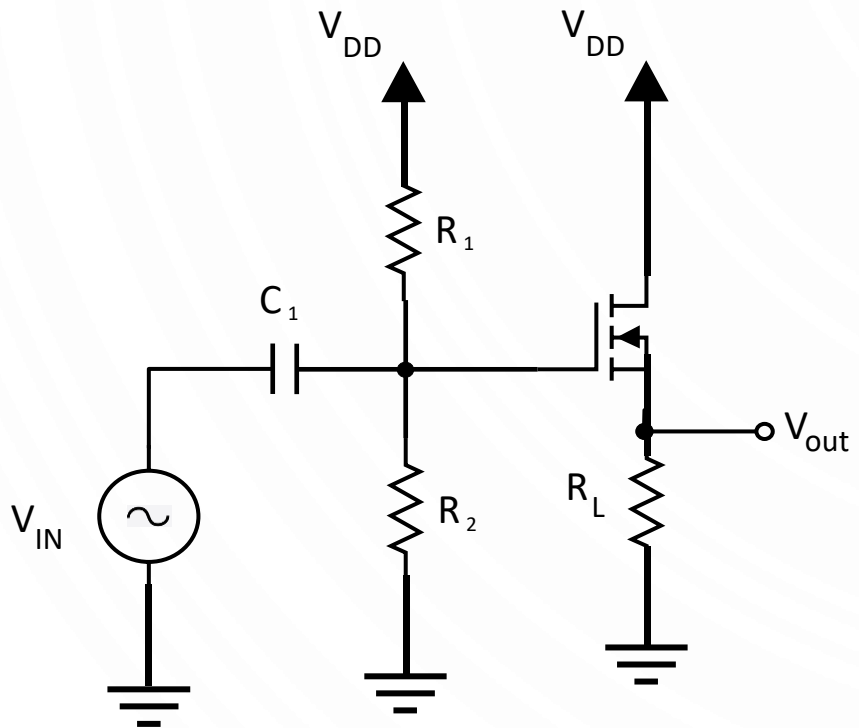
$$R_{in} = \frac{v_{in}}{i_{in}}$$

How to calculate output resistance:

1. Set all independent sources to zero (voltage sources $\rightarrow$ short circuit, current sources $\rightarrow$ open circuit).
2. Apply a test voltage v_x at the output.
3. Compute the resulting current i_x .
4. Take the ratio:

$$R_{out} = \frac{v_x}{i_x}$$

SINGLE STAGE AMPLIFIERS – COMMON DRAIN AMPLIFIER (SOURCE FOLLOWER)



$$A_v = \frac{V_{out}}{V_{in}} = \frac{g_m(R_{DS} || R_L)}{1 + g_m(R_{DS} || R_L)}$$

$$A_v = \frac{V_{out}}{V_{in}} = \frac{(R_{DS} || R_L)}{\frac{1}{g_m} + (R_{DS} || R_L)} \approx 1$$

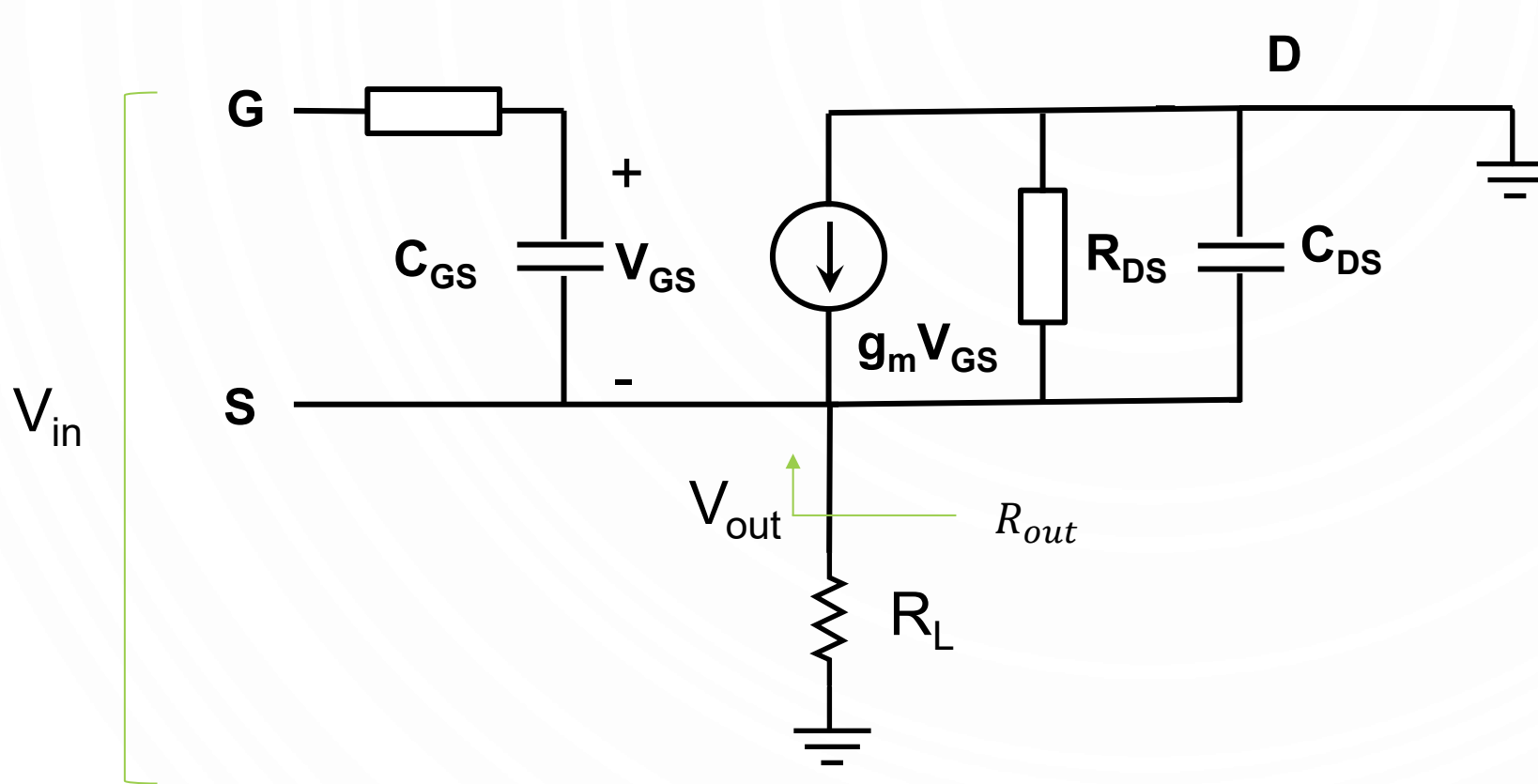
R_{DS} : output resistance of the MOSFET

$$A_v = \frac{V_{out}}{V_{in}} = \frac{g_m(R_{DS} || R_L)}{1 + g_m(R_{DS} || R_L)}$$

The voltage gain is positive!

The amplifier senses the input at the gate and produces the output at the source, with the drain tied to V_{DD} . V_{out} “follows” V_{in} . Since the dc level of V_{out} is lower than that of V_{in} by V_{GS} , we say the follower can serve as a “level shift” circuit. This topology exhibits a subunity gain.

SINGLE STAGE AMPLIFIERS – COMMON DRAIN AMPLIFIER (SOURCE FOLLOWER)



$$A_v = \frac{V_{out}}{V_{in}} = \frac{g_m(R_{DS} || R_L)}{1 + g_m(R_{DS} || R_L)}$$

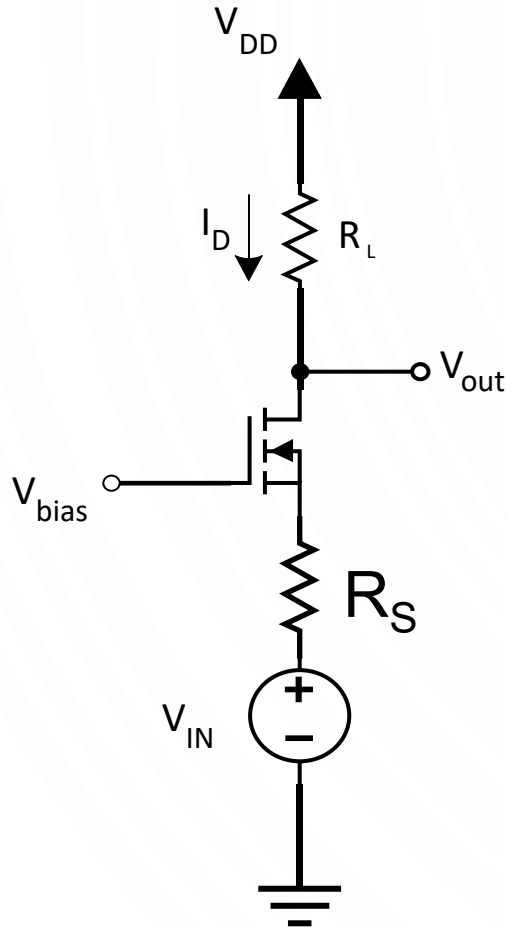
$$A_v = \frac{V_{out}}{V_{in}} = \frac{(R_{DS} || R_L)}{\frac{1}{g_m} + (R_{DS} || R_L)} \approx 1$$

$$R_{out} = R_L || 1/g_m$$

$$R_{in} = R_1 || R_2$$

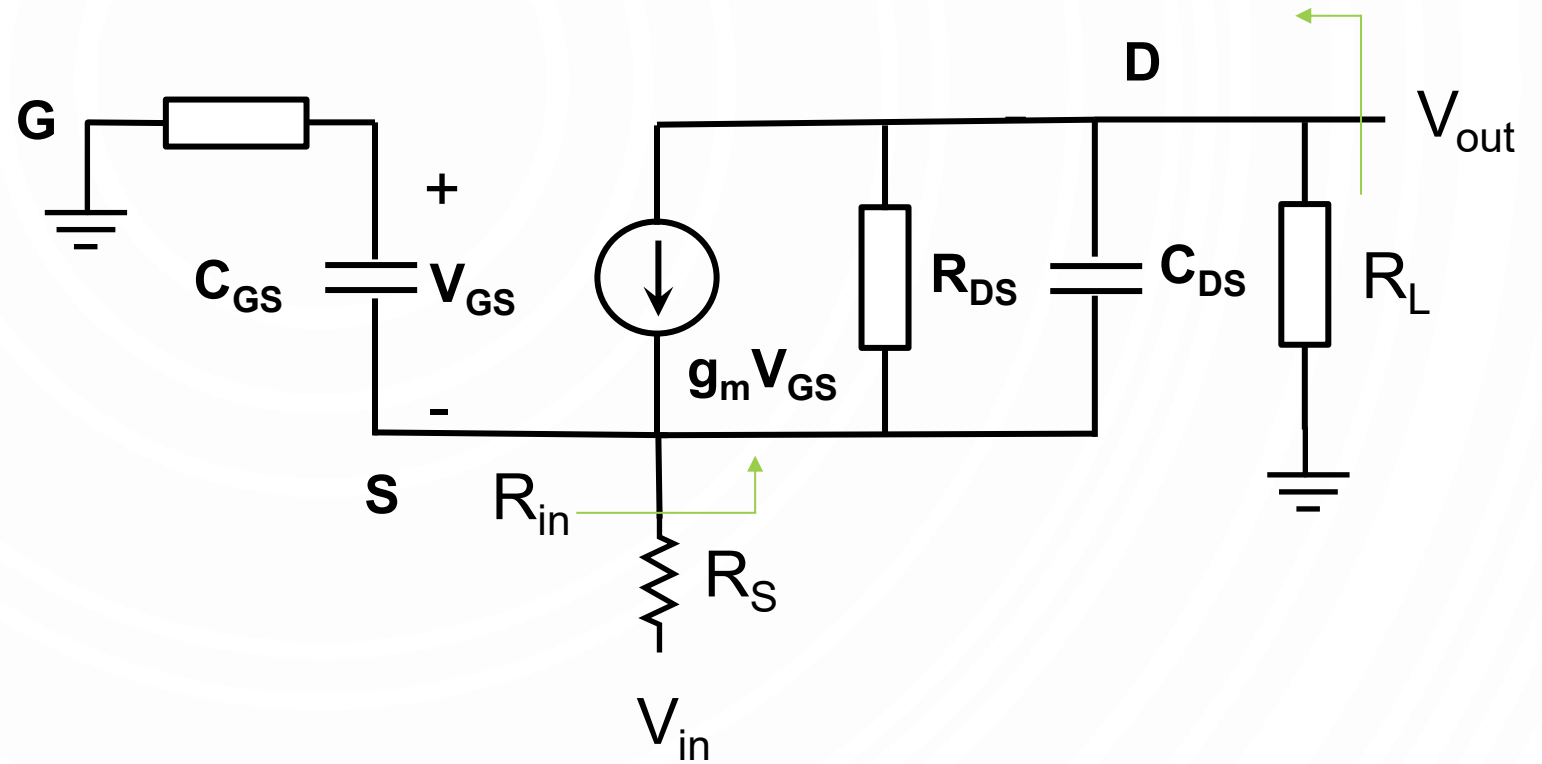
Without R_1 and R_2 , the input impedance of the amplifier is very large!

SINGLE STAGE AMPLIFIERS – COMMON GATE AMPLIFIER



$$A_v = g_m R_D$$

The voltage gain is positive but the input resistance is low, then lower gain compared to common-source! Why?



Try to derive R_{out} and g_m !



$$R_{out} \approx R_L || R_{DS} + R_S[1 + R_{DS}g_m]$$

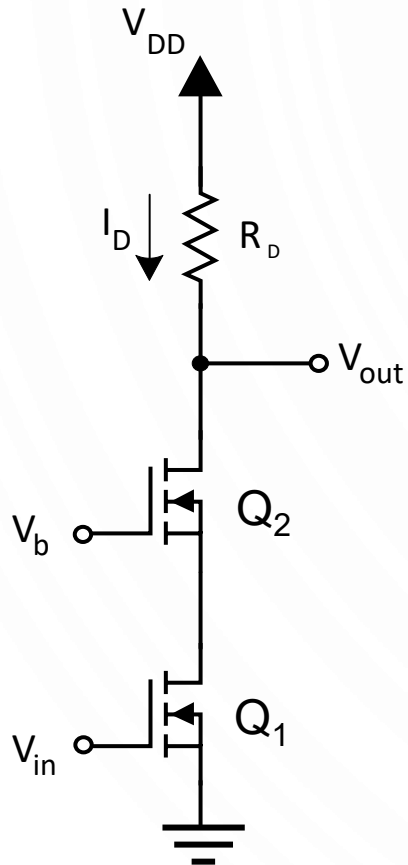
$$R_{in} \approx \frac{1}{g_m}$$

COMPARISON OF SINGLE STAGE AMPLIFIER TOPOLOGIES)

CONFIGURATION	COMMON GATE	COMMON DRAIN (SOURCE FOLLOWER)	COMMON SOURCE
Voltage gain	High	Low	High
Input resistance	Low	High	High
Output resistance	High	Low	Medium
Input / output phase relationship	0°	0°	180°

CASCODE STAGE

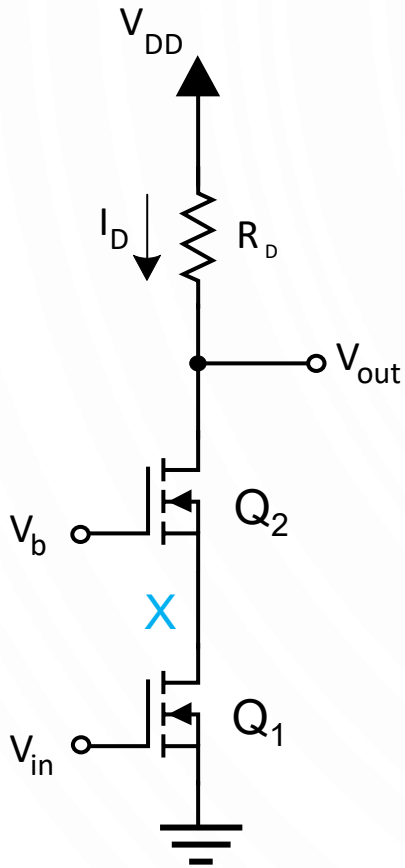
The cascade of a CS stage and a CG stage is called a cascode topology.



- Q_1 generates a small-signal drain current proportional to the small-signal input V_{in} and Q_2 simply routes the current to R_D .
- Q_1 is called the input device and Q_2 the cascode device.
- Q_1 and Q_2 in this example carry equal bias and signal currents.
- Both transistors operate in saturation region.

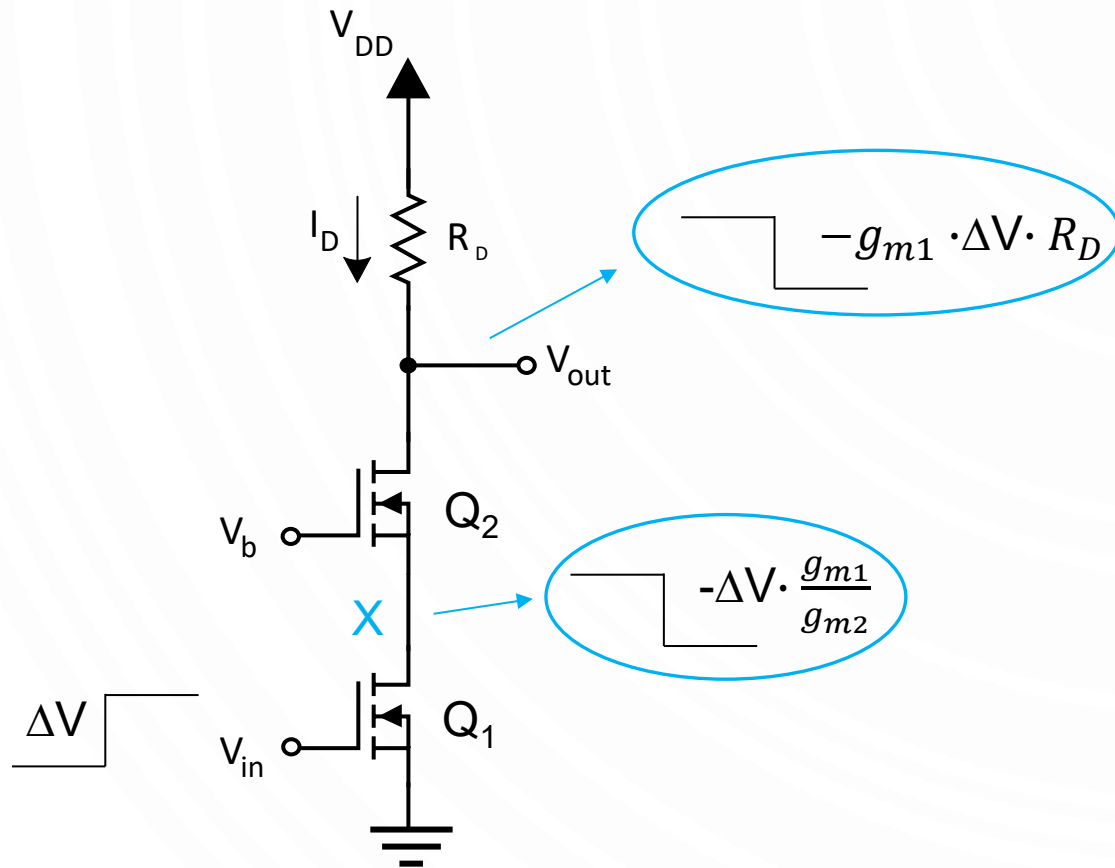
CS stage offers a large g_m and the CG stage has a low input impedance (can have a smaller width compared to the CS stage to lower the output capacitance) and a large output impedance, resulting in a large voltage gain.

ANALYSIS OF CASCODE STAGES



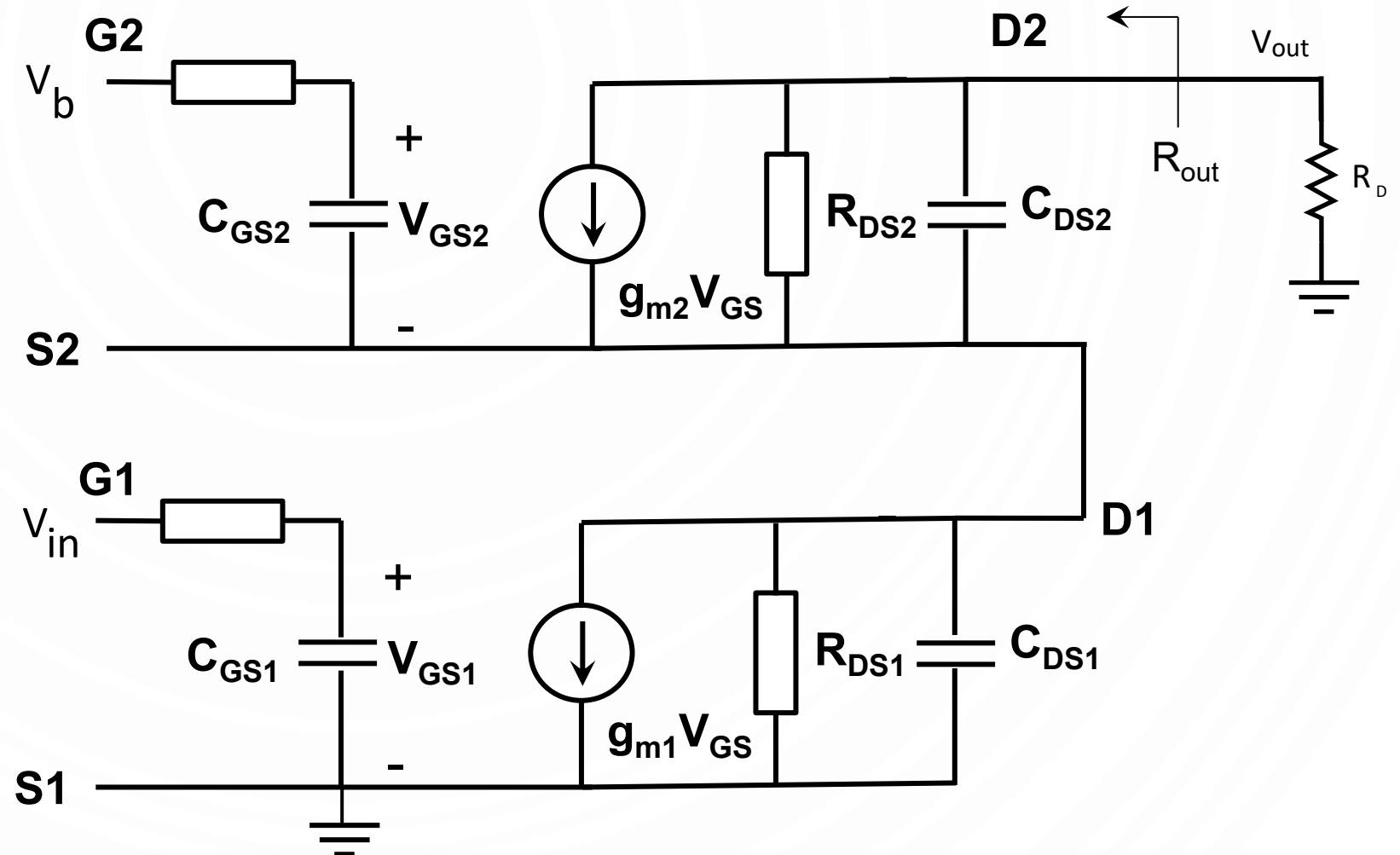
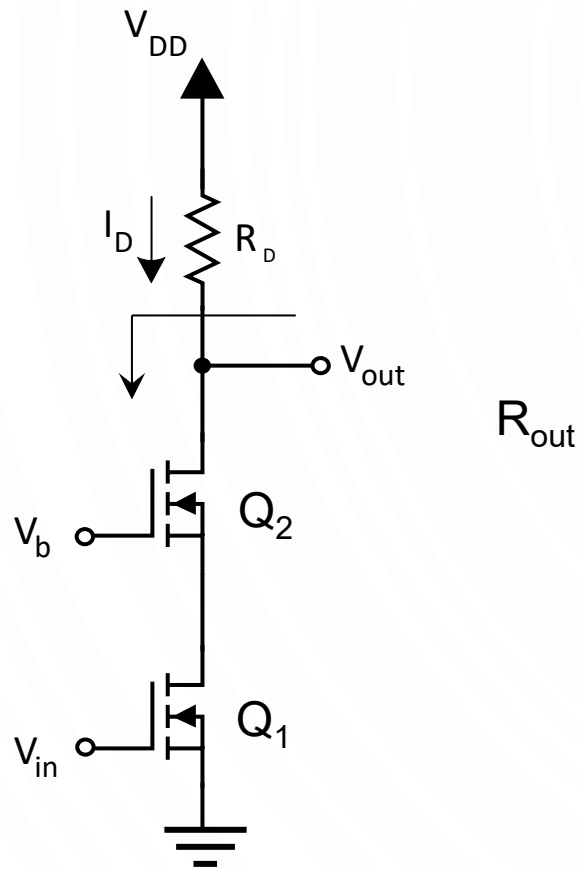
- For Q_1 to operate in saturation region, we must have
$$V_X \geq V_{in} - V_{th1}$$
- If both Q_1 and Q_2 are in saturation region, V_X is primarily determined by V_b :
$$V_b - V_{GS2} \geq V_{in} - V_{th1} \rightarrow V_b \geq V_{in} - V_{th1} + V_{GS2}$$
- For Q_2 to be saturated, $V_{out} \geq V_b - V_{th2}$, therefore
$$V_{out} \geq V_{in} - V_{th1} + V_{GS2} - V_{th2}$$
if V_b is chosen to place Q_1 at the edge of saturation.
- Addition of Q_2 to the circuit reduces the output voltage swing by at least its overdrive voltage.

ANALYSIS OF CASCODE STAGES



- Assume both transistors are in saturation and the channel length modulation is negligible.
- If V_{in} rises by ΔV , then I_{D1} increases by $g_{m1} \cdot \Delta V$.
- This change in current flows through the impedance seen at X or in other words at the source of Q_2 , which is equal $1/g_{m2}$.
- Thus, V_X falls by an amount given by $g_{m1} \cdot \Delta V \cdot 1/g_{m2}$.
- This change in I_{D1} also flows through R_D , producing a drop of $g_{m1} \cdot \Delta V \cdot R_D$ in V_{out} , just as in a simple CS stage.

ANALYSIS OF CASCODE STAGES



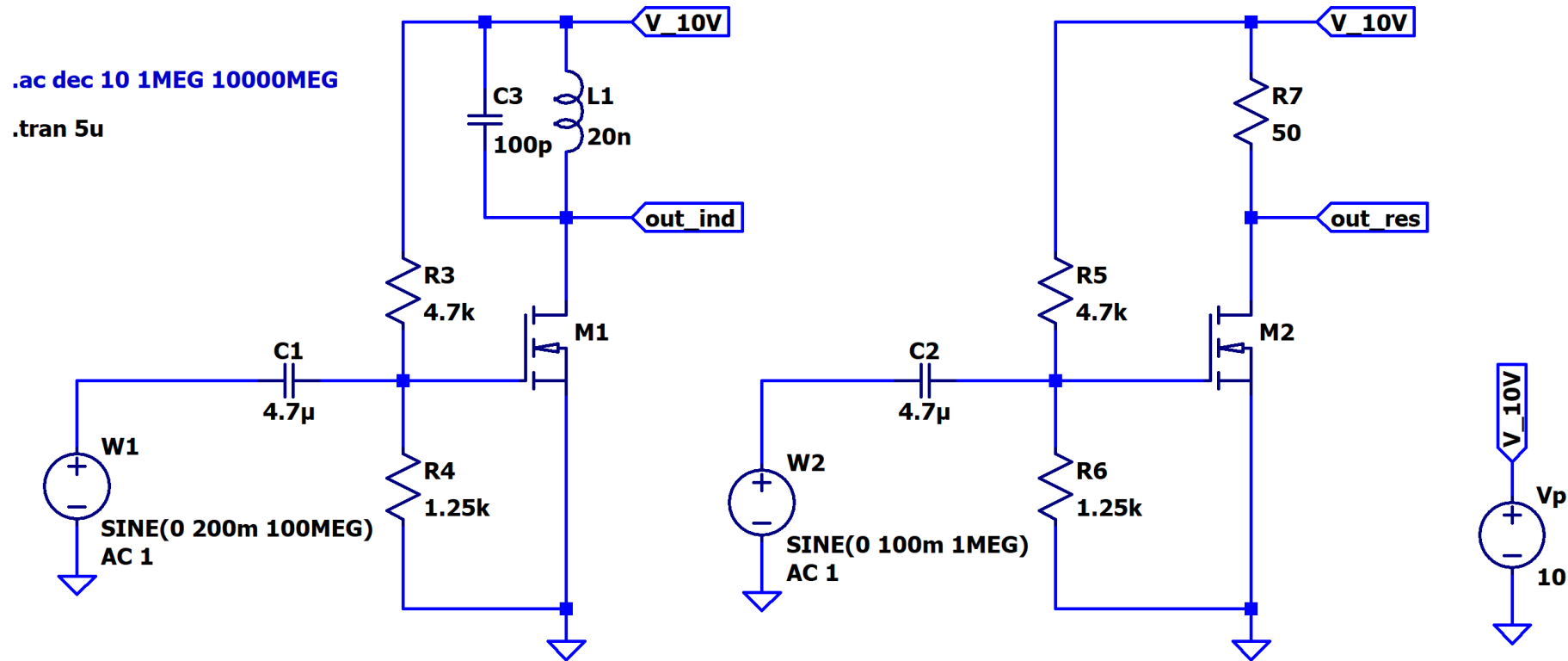
Important property of the cascode structure is its high output impedance:

$$R_{out} = [1 + g_{m1} R_{DS2}] R_{DS1} + R_{DS2}$$

EXAMPLES

Two common-source amplifiers are designed with resistive and inductive loads. Please determine the frequency at which the amplifiers deliver the maximum gain with AC analysis.

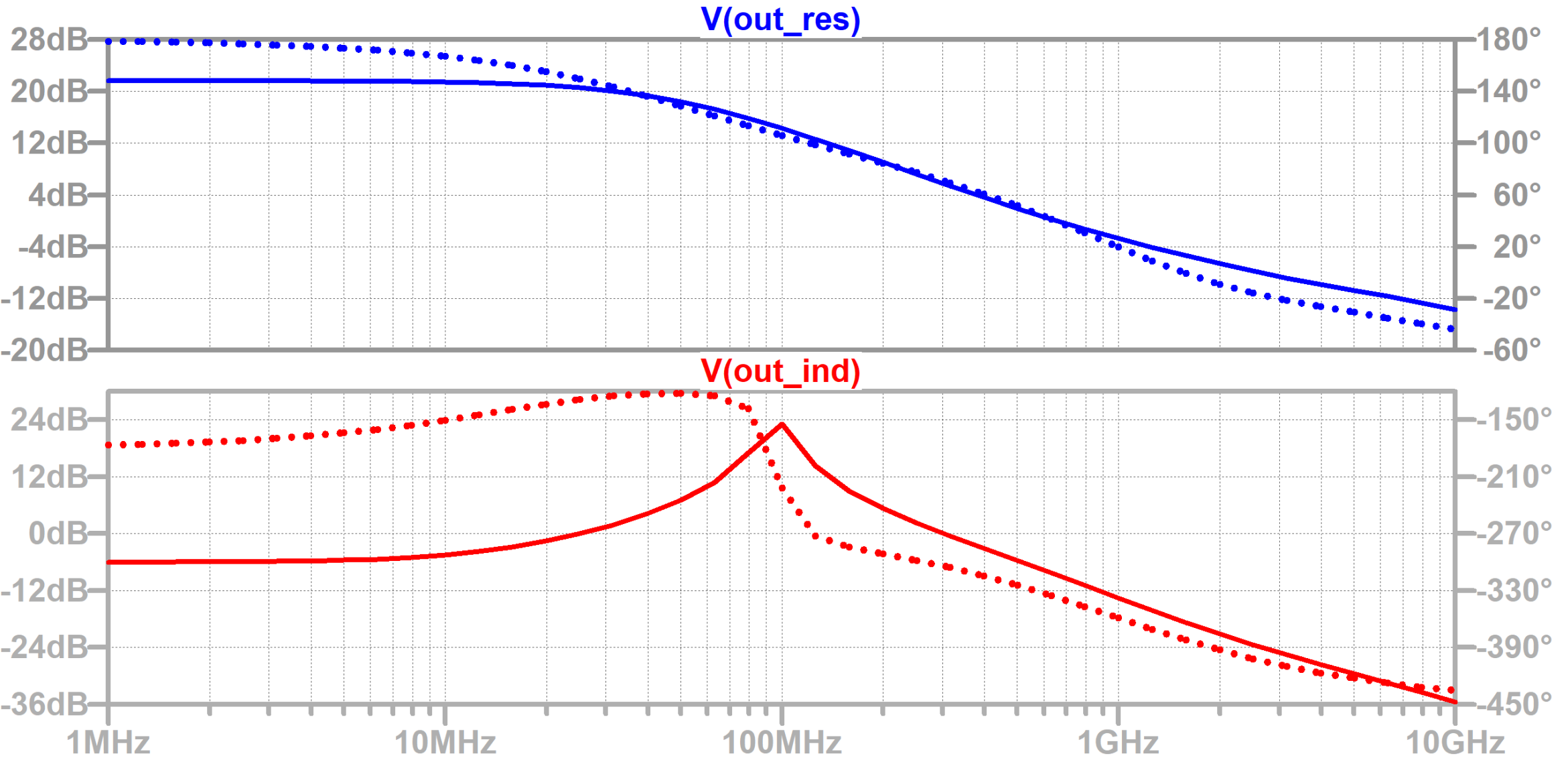
Observe the output voltage by means of a transient simulation.



Common-Source Amplifier with Inductive Load

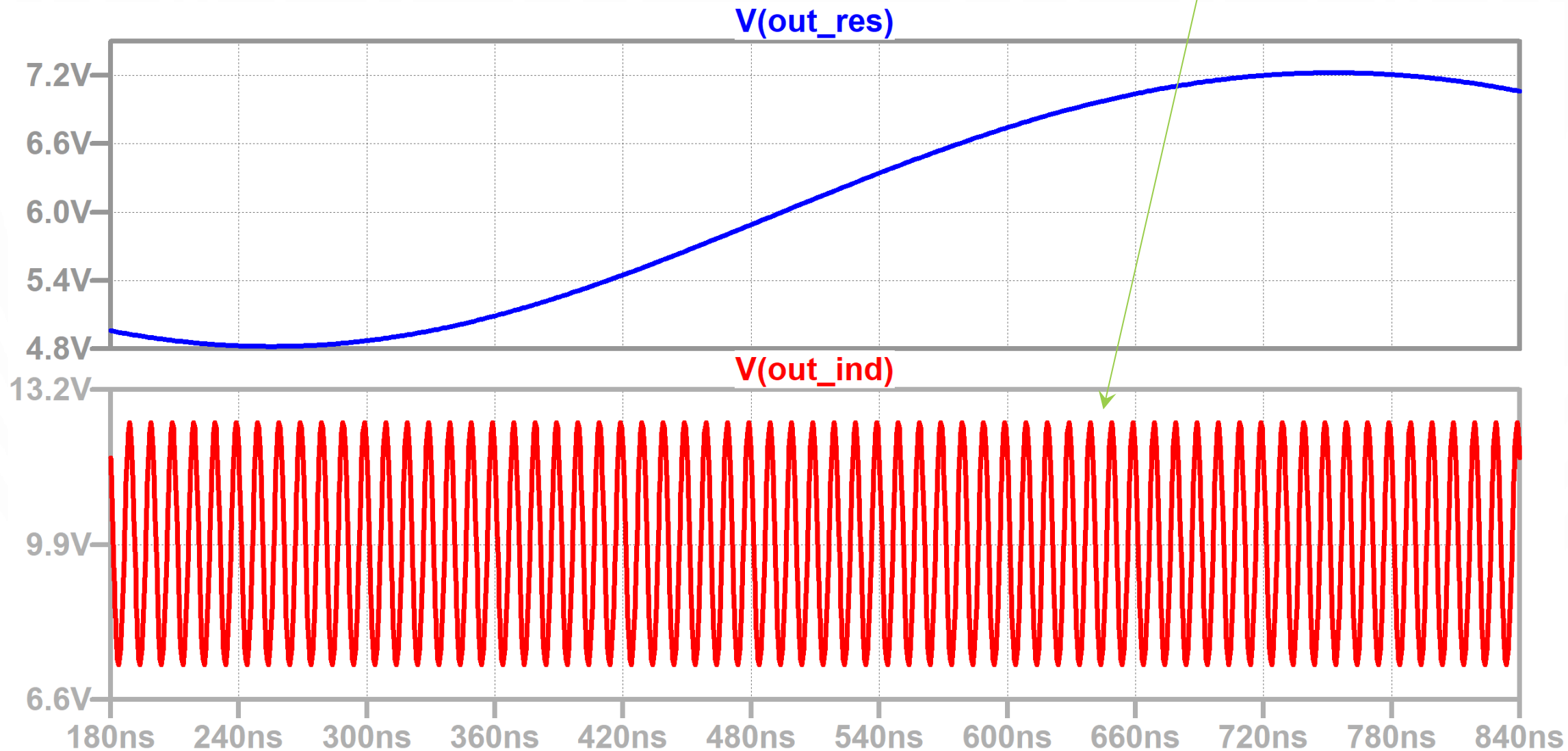
Common-Source Amplifier with Resistive Load

EXAMPLES



EXAMPLES

For inductive load, the output voltage is larger than the supply voltage!!!



References

1) www.halbleiter.org

The Principle of Semiconductor, Tokyo Electron Ltd.

2) Paul Gray, Robert Meyer, "Analysis and Design of Analog Integrated Circuits", John Wiley and Sons

3) Phillip E. Allen, Douglas R. Holberg, "CMOS Analog Circuit Design", Oxford University Press, 2002.